
Reference Code: 26670Q5E, Queue: AF2-424

General Information

Queue number

AF2-424

Project name

Shamrock Solar

Name of a signature authority from company

Zvi Tvizer

Title

Authorized Signatory

Project Developer (former Interconnection Customer)

Shamrock Solar, LLC

Federal Tax ID

39-4752369

Phone (day)

703-373-7345

Email of a signature authority from company

zvi.tvizer@energixrenewables.com

Country

United States

Business address

1201 Wilson Blvd, Suite 2200

City

Arlington

State

Virginia

Postal Code

22209

Primary Contact Name

Travis Bryant

Title

Senior Interconnection Specialist

Phone (day)

7033737345

Email

interconnection-us@energixrenewables.com

Invoice Contact Information

Invoice Contact Name

Billing Team

E-mail

ocrinvoices@energixrenewables.com

Phone (day)

703-373-7345

Facility Information

Facility location

GPS coordinates

Latitude

40.8626660

Longitude

-76.5913580

Is this a change in data to a previously submitted queue position or facility name?

No

Is this queue request claiming rights from a previous facility?

No

Planned in-service date

02.01.2027

Point of Interconnection

How many Points of Interconnection (POI) do you want to submit?

Primary

POI state

Pennsylvania

POI county

Northumberland

Connection type

Single line tap / New substation

Substation A (from)

Reed (REED - # 212377)

Substation B (to)

Reed Tap #1 (REED TP1 - # 212217)

Distance from POI to Substation A

0.85 Miles

Distance from POI to Substation B

7.01 Miles

Line voltage

69.000000 kV

Project Capability

Type of request

Build new facility

Describe the facility

40 MW of PV (AF2-424 and AF2-425 i.e Shamrock Solar, LLC) at POI solar facility consisting of 11 x SMA SC 4200-UP-US inverters each with its own step-up transformer connected to a single collector substation and stepped up to Transmission Voltage. Project will use trackers and thus will use CIRs = 0.6 x MFO

Generating Facility Capability

Maximum Facility Output (MFO) of the facility as defined in section 1.18A.03 of the PJM Tariff
40.000000 MW

Capacity Interconnection Rights (CIRs)
24.000000 MW

	Summer	Winter	
Net energy	40.000000	40.000000	MW
Gross energy output of facility	40.610000	40.610000	MW

Auxiliary Load

	Summer	Winter	
Total auxiliary load of facility	0.109700	0.109700	MW
Load related to the operations of the plant (e.g fans, pumps, etc)	0.036100	0.036100	MVAR

Where is the auxiliary load being connected?

High voltage side of the GSU

Station Load

Summer	Winter
--------	--------

Station load to support generation of electricity	0.000000	0.000000	MW
<i>Load necessary to support facility of the plant (e.g. heating, lighting, air-conditioning, office equipment etc)</i>			
	0.000000	0.000000	MVAR

Where is the station load being connected?

None

Total Reactive Power Capability at Max Gross Energy Output

	Summer	Winter	
Lagging - Overexcited	15.370000	15.370000	MVAR
Leading - Underexcited	-28.680000	-28.680000	MVAR

Additional comments related to the configuration above:

na

Fuel Supply Verification

Primary fuel type

Other

Behind The Meter

Purpose of application:

Not applicable

Describe electrical connections between the generation facility and the load.

na

Supporting Documents

Single line diagram

Shamrock Solar AF2-424 AF2-425, PJM NSA SLD, 2026-08-17.pdf

Comments

This project has 2 queue numbers associated with it (AF2-424 and AF2-425), studied to a total of 40 MW AC with a GIA that reflects this.

Generator Information

SMA 4200-UP-US - Solar

Number of identical inverter(s)

11

Specify manufacturer

SMA

Specify model

SC 4200-UP-US

MW value per inverter

3.691800 MW

MVA base

4.200000 MVA

Terminal voltage

0.630000 kV

Maximum fault current output from the inverter
The maximum fault current present 2 to 3 cycles after a fault; Typical range is 1.0 to 1.5 p.u.

1.340000 P.U.

Nominal power factor

0.950000 P.U.

Requested CIRs

24.00

Voltage relays ~~installed~~

~~Please see attached Dynamic Model Report~~

Frequency relays ~~installed~~

~~Please see attached Dynamic Model Report~~

Stability Models

Dynamic Model Package
Prepared in accordance with the [PJM Dynamic Model Development Guidelines](#)

Shamrock Solar AF2-424 AF2-425, PJM NSA ICF Dynamic Model, 2026-08-18.zip

Please provide a reactive capability P/Q curve for the inverter

Dynamic Model Report - Shamrock.pdf

Please upload the document showing frequency and voltage relay settings for the model

Dynamic Model Report - Shamrock.pdf

Transformer Information

SMA 4200 UP-US - Inverter Based Step-up Transformer

MVA base

4.200000 MVA

Number of identical transformers

11

Cooling class designation(s) for your transformer

How many ratings does the transformer have?

Two ratings

Rating 1

ONAN

4.200000 MVA

Rating 2

ONAF

4.200000 MVA

Winding Type

Two Windings

Impedance on MVA base

	R	X	X/R
High-side to low-side	0.008458 P.U.	0.084578 P.U.	9.999764

	RO	X0	X/R
High-side to low-side	0.008458 P.U.	0.084578 P.U.	9.999764

Winding Nominal Voltages

High-side

34.500000 kV

Low-side

0.630000 kV

Winding Connection Types

High-side

Delta

Low-side

Wye

Tap position

3.000000

Number of taps

5

Step size

2.500000

MPT - Inverter Based Main Transformer

MVA base

45.000000 MVA

Number of identical transformers

1

Cooling class designation(s) for your transformer

How many ratings does the transformer have?

Three ratings

Rating 1

ONAN

45.000000 MVA

Rating 2

ONAF

60.000000 MVA

Rating 3

ONAF

75.000000 MVA

Winding Type

Two Windings

Impedance on MVA base

	R	X	X/R
High-side to low-side	0.001802 P.U.	0.073378 P.U.	40.720311
	R0	X0	X/R
High-side to low-side	0.001622 P.U.	0.066040 P.U.	40.715166

Short
Circuit
Study: 2:
The zero-
sequence
impedance
values of
the MPT on
QP
(Z0=0.06605
p.u) does
not match
the
impedance
values
stated in
the FAT
report
(Z0=0.0692
p.u).
During the
study the
value from
the FAT will
be
utilized. -
WORKNA
[Remove](#)

Winding Nominal Voltages

High-side

69.000000 kV

Low-side

34.500000 kV

Winding Connection Types

High-side

Grounded Wye

Low-side

Delta

Tap position

17.000000

Number of taps

33

Step size

0.625000

Collector System Equivalent

Does the project have collector system equivalent data (modeled between main transformer and step-up transformers)?

Yes

Voltage level

34.500000 kV

MVA base

100.000000 MVA

Impedance on 100 MVA Base

	R	X	
Total branch positive sequence impedance	0.009811	0.006044	P.U.
Total branch zero sequence impedance	0.029803	0.009738	P.U.
		B	
Total branch charging susceptance		0.006628	P.U.

Collector System Equivalent data

Attachment Line Data

Voltage level

69.000000 kV

MVA base

100.000000 MVA

Attachment line length

0.02 Miles

Conductor type

ACSR

Impedance on 100 MVA Base

	R	X	
Total branch positive sequence impedance	0.000128	0.000295	P.U.
Total branch zero sequence impedance	0.000385	0.000885	P.U.
		B	
Total branch charging susceptance		0.000005	P.U.

Readiness Form

Maximum Facility Output (MFO)

Capacity Interconnection Rights (CIR)

Megawatt energy value (MWE)

Readiness Deposit One

Payment Type

Supporting Documents

Deficiency Review

TO Comments

Additional comments to Interconnection Customer

B

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U

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Sans Serif

Normal

1

2

3

☰

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08/06/2026 (AW): Load flow data is clear.

08/07/2026 (XR): Short circuit data is deficient. Please address the comments above.

08/07/2026 (XR): Stability data is deficient. Please address the comments above.

08/24/2026 (AW): Load flow data is clear.

08/24/2026 (MS): Stability data is clear.

08/24/2026 (MS): Short circuit data is clear.

08/24/2026 (AW): Attachment # 1 is acceptable.

Notes for PJM internal purposes

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Sans Serif

Normal

1

2

3

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LSAT Uploads (Attorney-Client Privilege)

Reviewing as Load Flow Engineer

Save and Preview

Deficient

Review Complete

Back